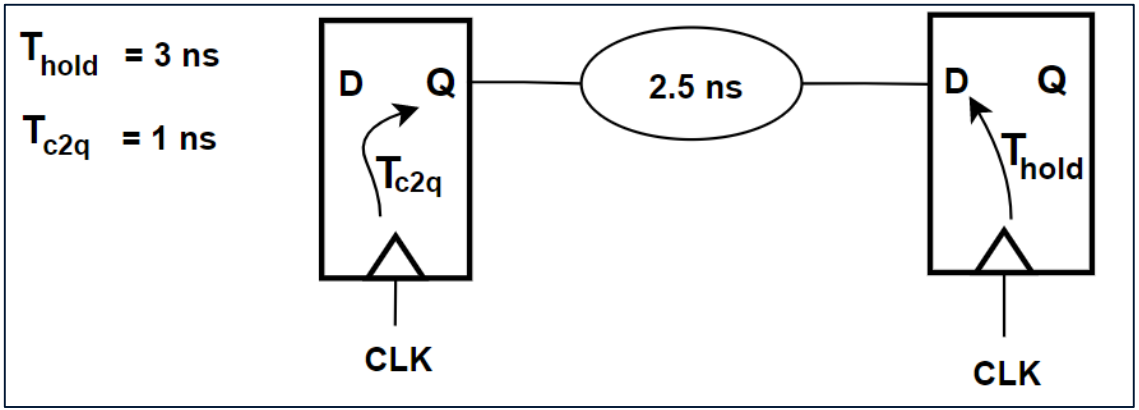
Example 1

- **Assumptions**

1. Interconnect Delays not considered
2. Clock Uncertainty not considered
3. One Operating Condition

- **Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack



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Example 2

- **Assumptions**

1. Interconnect Delays not considered
2. **Clock Uncertainty is considered**
3. One Operating Condition

- **Required**

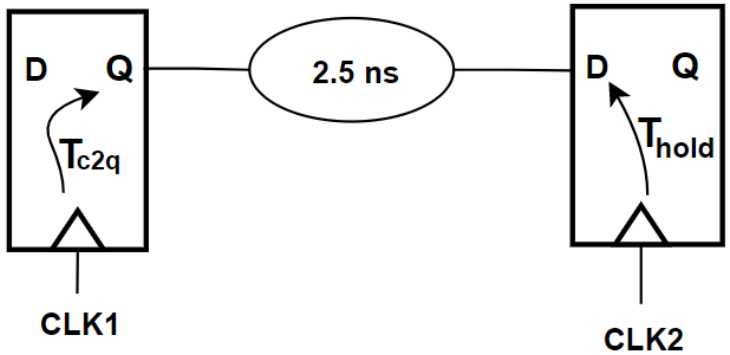
1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack

T_{hold} = 3 ns

T_{c2q} = 1 ns

Skew

CLK 1_2 = 0.7 ns



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Example 3

- **Assumptions**

1. **Interconnect Delays is considered**
2. **Clock Uncertainty is considered**
3. One Operating Condition

- **Required**

1. Is this path timing is met or not?

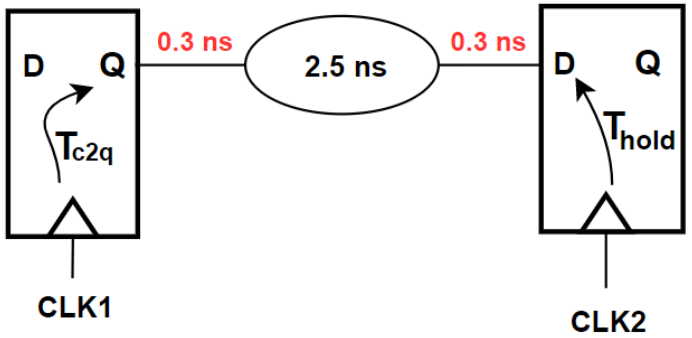
2. if met calculate the positive slack and if not calculate the negative slack

T_{hold} = 3 ns

T_{c2q} = 1 ns

Skew

CLK 1_2 = 0.7 ns



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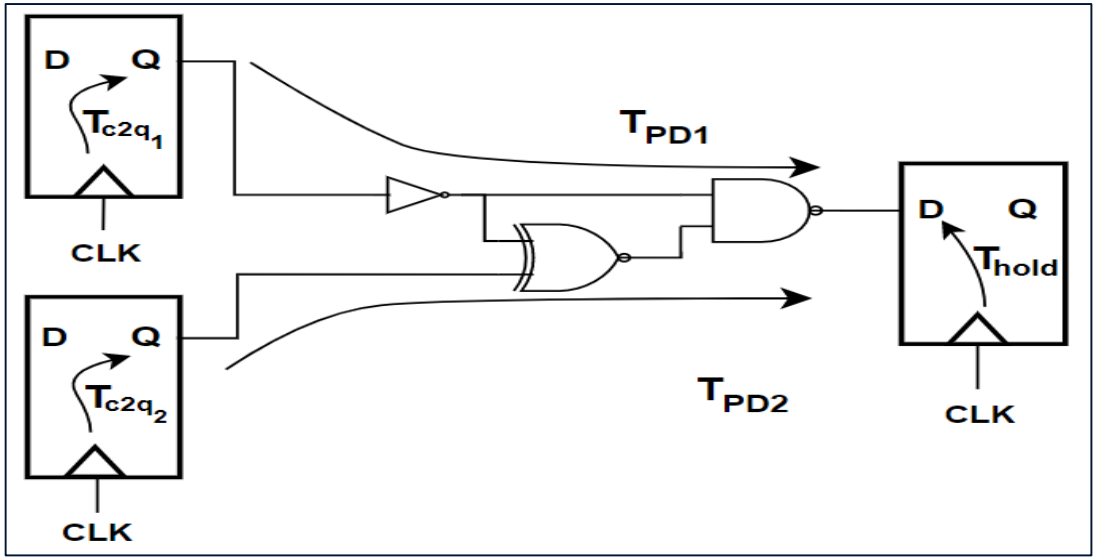
Example 4

- **Assumptions**

1. Interconnect Delays is not considered
2. Clock Uncertainty is not considered
3. **Worst/Best Operating Conditions**

- **Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack
3. if not met calculate the negative slack and suggest a solution to fix hold violation



Delays	Inverter	NOR	NAND	Tcq1	Tcq2	Thold
Max	0.5	1.5	1	0.4	0.35	2.2
Min	0.45	1.3	0.95	0.3	0.3	2

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